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Studierichting: Informatica
Oefeningenreeks 7C nr. 2.3

$f(x, y) = e^{(x-2y^2)^2}$

Eerste orde:

$$\frac{\partial f}{\partial x} = e^{(x-2y^2)^2} \cdot (2x - 4y^2)$$
$$\frac{\partial f}{\partial y} = e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)$$

Tweede orde:

$$\frac{\partial^2 f}{\partial x^2} = e^{(x-2y^2)^2} \cdot (2x - 4y^2)^2 + 2e^{(x-2y^2)^2}$$
$$= 2e^{(x-2y^2)^2} \cdot \big(2(x - 2y^2)^2 + 1\big)$$

$$\frac{\partial^2 f}{\partial y^2} = e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)^2 + e^{(x-2y^2)^2} \cdot (-8xy + 48y^2)$$
$$= 8e^{(x-2y^2)^2} \cdot \big(8(-xy + 2y^3)^2 - x + 6y^2\big)$$

$$\frac{\partial^2 f}{\partial x \partial y} = e^{(x-2y^2)^2} \cdot (2x + 4y^2)^2 + e^{(x-2y^2)^2} \cdot (-8y)$$
$$= 4e^{(x-2y^2)^2} \cdot \big((x - 2y^2)^2 - 2y\big)$$

$$\frac{\partial^2 f}{\partial y \partial x} = e^{(x-2y^2)^2} \cdot (-8xy + 16y^3)^2 + e^{(x-2y^2)^2} \cdot (-8y)$$
$$= 8e^{(x-2y^2)^2} \cdot \big(8(-xy - 2y^3)^2 - y\big)$$

References